

Cloud & IoT Security

1. Introduction

Cloud & IoT Security focuses on protecting data, devices, and systems in **cloud environments** and the **Internet of Things (IoT)**.

Both technologies are widely used today — cloud for **data storage and computing**, and IoT for **smart devices** like sensors, cameras, and wearables.

Because they are **connected to the internet**, both are vulnerable to **hacking, data breaches, and unauthorized access**, making strong security essential.

2. Cloud Security

a. What is Cloud Security?

Cloud Security refers to the set of **policies, technologies, and controls** used to protect **data, applications, and infrastructure** hosted in cloud environments.

It ensures the **confidentiality, integrity, and availability (CIA)** of cloud resources.

b. Major Cloud Security Threats

Threat	Description
Data Breaches	Unauthorized access to sensitive data stored in the cloud.

Data Loss	Accidental deletion or hardware failure causing permanent data loss.
Account Hijacking	Attackers steal cloud credentials (like AWS, Azure, or Google Cloud accounts).
Insecure APIs	Poorly secured application interfaces can be exploited.
Denial of Service (DoS)	Overloading cloud services to make them unavailable.
Misconfiguration	Weak permissions or open storage buckets (e.g., AWS S3 leaks).

c. Cloud Security Models

Cloud security is a **shared responsibility** between the **Cloud Service Provider (CSP)** and the **Customer**.

Cloud Model	CSP Responsibility	Customer Responsibility
IaaS (Infrastructure as a Service)	Hardware, storage, and network security	OS, apps, and data security
PaaS (Platform as a Service)	Platform and runtime security	Application and user data
SaaS (Software as a Service)	Entire app and infrastructure	User access and data management

Example:

In Google Drive, Google secures the system; users must use strong passwords and avoid sharing links publicly.

d. Cloud Security Best Practices

- Use **strong authentication** (MFA).
- Encrypt all stored and transmitted data.
- Keep backups in different regions.
- Choose trusted and compliant cloud providers.

3. IoT Security

a. What is IoT Security?

Internet of Things (IoT) refers to the network of **connected smart devices** (like sensors, thermostats, smart TVs, CCTV cameras, etc.) that collect and share data.

IoT Security protects these devices and the networks they operate on from cyber threats.

b. IoT Security Challenges

Challenge	Description
Weak Authentication	Default or simple passwords can be easily hacked.
Unpatched Devices	Many IoT devices don't get regular software updates.
Data Privacy Risks	Devices collect personal data that can be misused.
Botnet Attacks	Infected IoT devices can be used in large-scale attacks (e.g., Mirai Botnet).
Lack of Encryption	Data sent between IoT devices is often unencrypted.

Physical Tampering	Devices placed in public areas can be physically accessed or modified.
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c. Common IoT Security Protocols

Protocol	Purpose	Example Use
MQTT (Message Queuing Telemetry Transport)	Lightweight communication protocol	Smart sensors
CoAP (Constrained Application Protocol)	Used for low-power IoT devices	Smart home systems
DTLS (Datagram Transport Layer Security)	Secures UDP-based IoT communication	Industrial IoT networks

d. Best Practices for IoT Security

- Change **default passwords** immediately.
- Keep **firmware** up to date.
- Use **strong encryption (AES, TLS)**.
- Regularly monitor device activity.

4. Difference Between Cloud and IoT Security

Aspect	Cloud Security	IoT Security
Focus	Protecting data, apps, and infrastructure in cloud systems	Protecting smart devices and their data
Main Threats	Data breaches, insecure APIs, misconfigurations	Weak passwords, botnets, firmware attacks

Main Mechanisms	Encryption, IAM, Firewalls, MFA	Secure boot, firmware updates, device authentication
Responsibility	Shared between provider and user	Mainly user or manufacturer responsibility
Example	Securing Google Cloud data	Securing a smart home or CCTV network

